

UŽITOČNÉ GONIOMETRICKÉ VZORČEKY

$\sin$ a $\cos$ sú tzv. harmonické funkcie.

1. $\sin$ a $\cos$ sú periodické s (uhlovou) periódou 2π :

$$\begin{aligned}\sin(\heartsuit) &= \sin(\heartsuit + 2\pi) = \sin(\heartsuit + 4\pi) = \dots \\ &= \sin(\heartsuit - 2\pi) = \sin(\heartsuit - 4\pi) = \dots \\ \cos(\heartsuit) &= \cos(\heartsuit + 2\pi) = \cos(\heartsuit + 4\pi) = \dots \\ &= \cos(\heartsuit - 2\pi) = \cos(\heartsuit - 4\pi) = \dots\end{aligned}\tag{1}$$

2. $\sin$ je nepárna funkcia, $\cos$ je párna funkcia:

$$\begin{aligned}\sin(-\heartsuit) &= -\sin(\heartsuit) \\ \cos(-\heartsuit) &= \cos(\heartsuit)\end{aligned}\tag{2}$$

3. $\sin$ a $\cos$ sú vzájomne (fázovo) posunuté o uhol $\pi/2$:

$$\begin{aligned}\sin(\heartsuit) &= \cos\left(\heartsuit - \frac{\pi}{2}\right) \\ \cos(\heartsuit) &= \sin\left(\heartsuit + \frac{\pi}{2}\right)\end{aligned}\tag{3}$$

4. Pytagorova veta

$$\sin^2(\heartsuit) + \cos^2(\heartsuit) = 1\tag{4}$$

bez ohľadu na $\heartsuit$!

5. „rozkladacie“ vzorčky

$$\begin{aligned}\sin(\heartsuit \pm \diamondsuit) &= \sin(\heartsuit) \cdot \cos(\diamondsuit) \pm \cos(\heartsuit) \cdot \sin(\diamondsuit) \\ \cos(\heartsuit \pm \diamondsuit) &= \cos(\heartsuit) \cdot \cos(\diamondsuit) \mp \sin(\heartsuit) \cdot \sin(\diamondsuit)\end{aligned}\tag{5}$$

6. $\sin$ a $\cos$ dvojnásobných uhlov:

$$\begin{aligned}\cos(2\heartsuit) &= \cos^2(\heartsuit) - \sin^2(\heartsuit) \\ \sin(2\heartsuit) &= 2 \sin(\heartsuit) \cdot \cos(\heartsuit)\end{aligned}\tag{6}$$

7. definícia funkcií $\tan$ a $\cot$:

$$\begin{aligned}\tan(\heartsuit) &= \frac{\sin(\heartsuit)}{\cos(\heartsuit)} \\ \cot(\heartsuit) &= \frac{\cos(\heartsuit)}{\sin(\heartsuit)}\end{aligned}\tag{7}$$

8. Goniometrické funkcie malých uhlov $\varphi \approx 0$:

$$\begin{aligned}\sin(\varphi) &\approx \varphi \\ \cos(\varphi) &\approx 1 \\ \tan(\varphi) &\approx \varphi\end{aligned}\tag{8}$$